For a plane circular probe contact, one finds a total resistance R of:

$$R = \frac{\rho}{4a} \quad . \tag{14}$$

Approximately 80% of the potential drop occurs within a distance of 5a from the probe. This means that one essentially measures the resistance of a very small volume of material. There are a number of practical problems associated with the spreading resistance probe. It is difficult to find hard metals or alloys with high conductivity. It also is difficult to keep lateral motions of the probe below 1 μm. Silicon wafer manufacturers usually screen their wafers for conductivity fluctuations due to dynamical Impurity segregation effects during crystal growth.

Hall Effect

The techniques discussed so far allow the measurement of the conductivity of materials with a variety of geometries and fluctuations down to a scale of a few μm. In the cases in which. either the actual carrier concentration or the mobility needs to be known, one usually uses a Hall effect measurement; This effect is a direct consequence **of** the Lorentz force, the force a moving charge carrier feels In a magnetic field.

The Lorentz force $\vec{F}$ is:

$$\vec{F} = e\,(\vec{v} \times \vec{B}) \tag{15}$$

In the classical Hall effect geometry, the three vectors $\vec{F}$, $\vec{v}$ and $\vec{B}$ are chosen at right angles. $\vec{F}$ leads to a build up of a surplus of charges on one side of the conductor and to a deficit on the other side. This

charge separation creates an electric field and is integrated to the Hall voltage V_H .

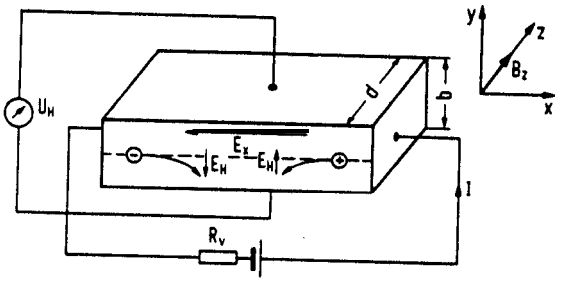


Figure 7

Eq. (15) can be rewritten for the special case shown in Fig. 7:

$$F_y = e\ V_x \cdot B_z = e\ E_{Hall} \qquad (16)$$

The current density $j_x = e\ v_x\ c$ with c = charge carrier concentrations. Eq. (16) is often written in the following form:

$$E_{Hall} = R_{Hall} \cdot j \cdot B \qquad (17)$$

with R_{Hall} = Hall coefficient $= \frac{1}{e \cdot c}$ (18)

Applying the cork screw rule, we find that for both types of carrier charges, negative as well as positive, the Lorentz force points in the same direction. (Try it out but do not forget to change to sign of e, the charge of the electron in equation (15) when switching from positive to negative charges!) This means that we can determine the sign of the moving charge carriers as well as their concentration!

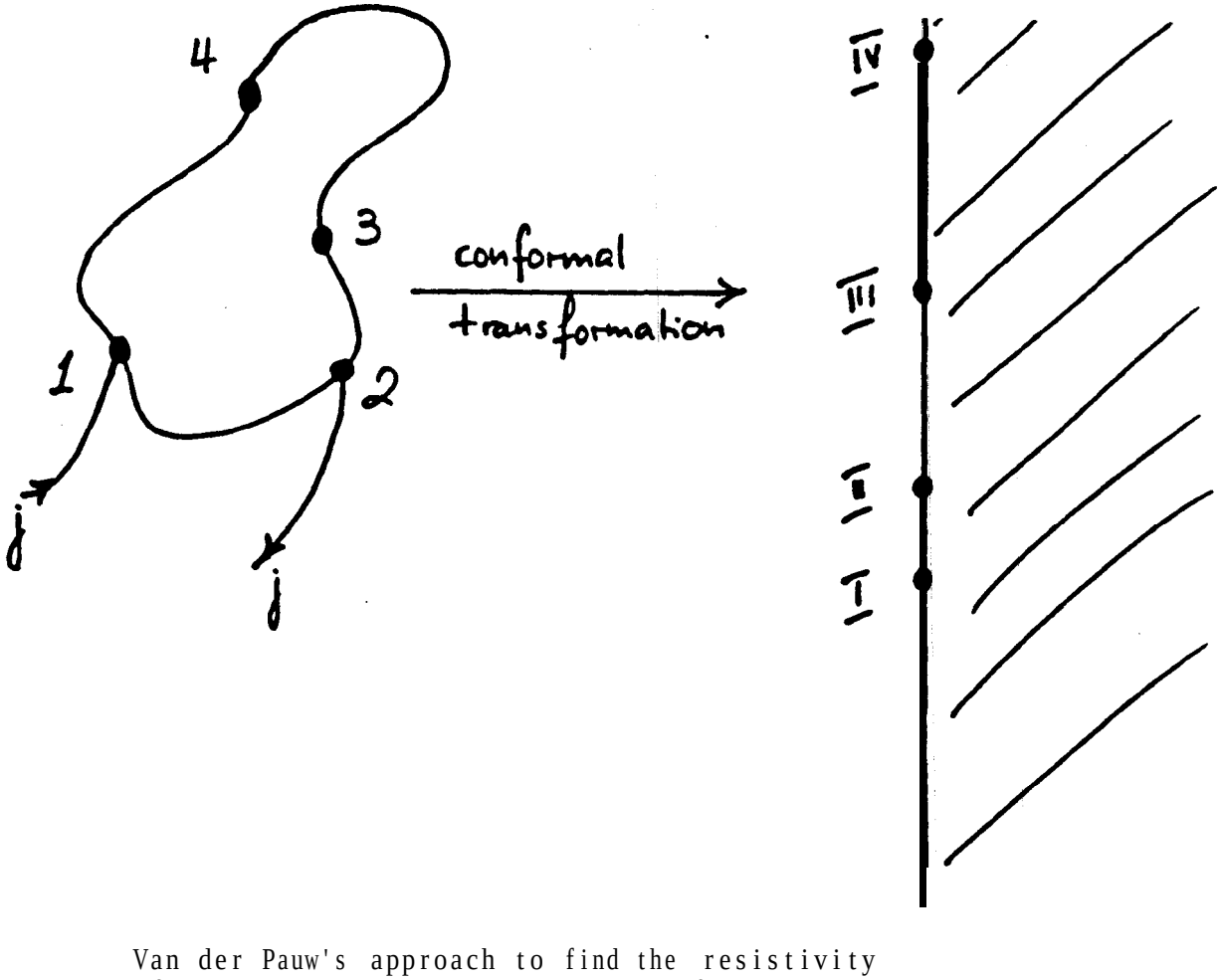
For the general case in which both kinds of carriers are present (e.g. intrinsic semiconductors) eq. (18) can be generalized to:

$$R_H = \frac{1}{e} \frac{p - b^2 n}{(p + bn)^2}$$

with p = positive carrier concentration, n = negative carrier concentration, b= ratio of negative to positive carrier mobility = μ_n/μ_p.

Van der Pauw Hall Effect Geometry

In order to obtain accurate results with the standard parallelpiped Hall sample, one has to have excellent control of the sample shape and perfect electrical contacts; Neither of the two demands are always easily fulfilled; Van der Pauw (Ref. Fig. 9) suggested a sample geometry which does not have to be precisely known but can still give highly accurate results; He showed that conformal mapping (conformal transformation) can be used to image any singly connected thin sheet of uniformly conducting material with four point contacts on its periphery on a half plane. The contacts will lie on the half plane border.



Van der Pauw's approach to find the resistivity of an arbitrarily shaped lamellae of thickness δ.

Figure 8

The current flow pattern and equipotential lines are easily calculated in the half plane. One can futher show that the resistivity and Hall coefficient can be obtained from

$$\rho = \frac{\pi \delta}{\ell n 2} \cdot \frac{R_{12,34} + R_{23,41}}{2} \cdot f \qquad (20)$$

with

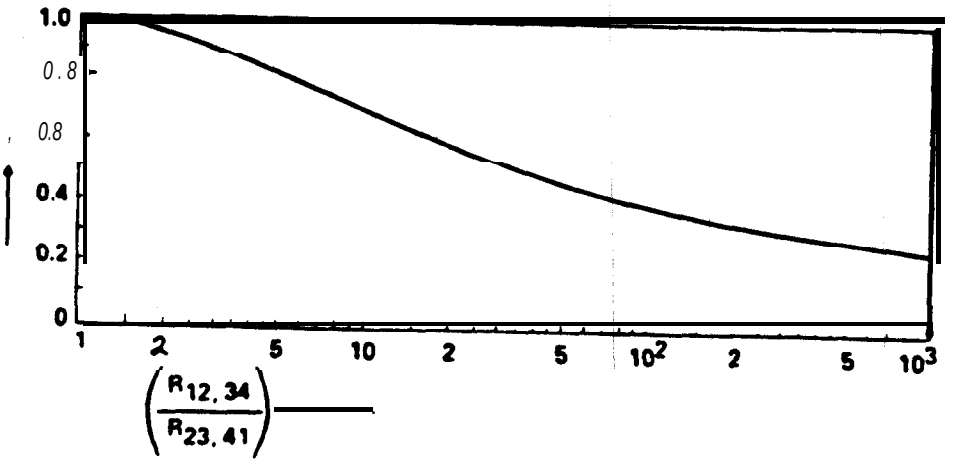
δ = thickness of the conductor

$$R_{12,34} = \Delta V_{34}/I_{12} \qquad (21)$$

$$R_{23,41} = \Delta V_{41}/I_{23}$$

etc.

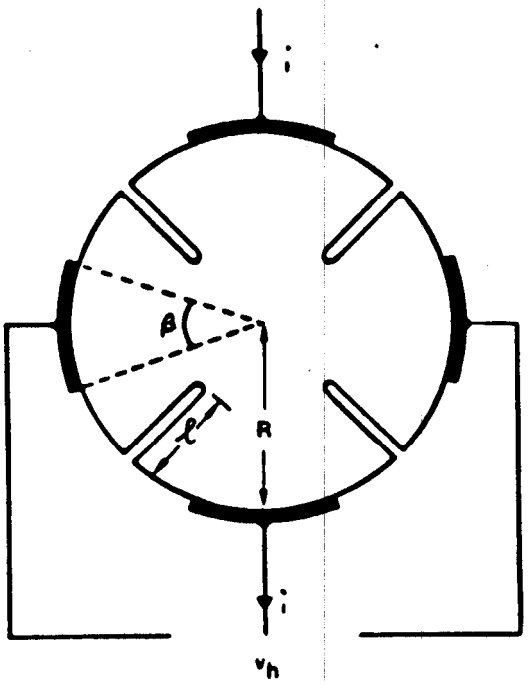
The factor f is slowly varying with the ratio of $R_{12,34}/R_{23,41}$.



The function f used to determine the specific **resistivity** of a specimen with an arbitrary contour as a function **of** the resistance ratio $R_{12,34}/R_{23,41}$. (after L. J. van der Pauw, **Phillips Rcs. Repts., 13 (1958) 1).**

Figure 9

One of the most commonly chosen shapes is the circle. The conductance of and free carrier concentration in a thin implanted layer of silicon is determined by many Van der Pauw '"clover leaves" using photomasking techniques. The etching is done to a depth which guarantees full isolation of individual "clover leaves" from each other.



Clover leaf-shaped structure for Hall effect measurements shows slots **of** length ***l*** **cut** into disc of radius R. **Contacts subtend angle β** (after **E. Jacobs and C. de Mey.** Electron. Lett., 12 (1976) 4).

Figure 10

The Hall coefficient RH is simply

$$R_H = \frac{\delta \, \Delta R_{24,13}}{|\vec{B}|} \tag{22}$$

with $\Delta R_{24,13} = \Delta V_{13}/I_{24}$.

ΔV_{13} corresponds to the usual Hall voltage.

It is the difference between the voltages produced across the electrodes 1 and 3 with and without a B-field that determines the Hall coefficient. $\vec{B}$ is chosen perpendicular to the sample under investigation. Equation (22) can be rewritten to solve for the free carrier concentration c:

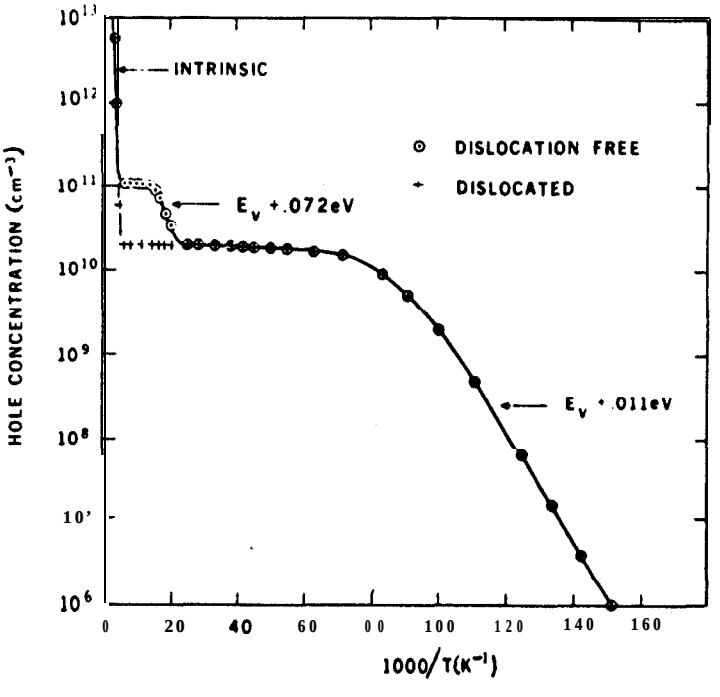
$$c = \frac{|\vec{B}|}{\delta \, \Delta R_{24,13} \, e} \qquad (23)$$

Eqns. (20) and (23) combined lead to the mobility μ:

$$\mu = \frac{\delta \, \Delta R_{24,13}}{|\vec{B}| \cdot g} \qquad (24)$$

Variable Temperature Hall Effect in Semiconductors

Hall effect measurements can provide valuable information about the size of the bandgap and the location and concentration of defect or impurity levels inside the bandgap of insulators and semiconductors. As a representative example we will discuss ultrapure germanium, both dislocation free and with dislocations (Fig. 6).



Arrhenius plot of the free hole concentration p (log p versus 1000/T) The dislocation-free sample contains an acceptor level at $E_V + 72$ meV.

Figure 11

a. Ge Single Crystal with Dislocations (crosses)

In the so-called intrinsic range, the hole and electron concentrations are equal and are given' by band-band transitions across the band gap. The intrinsic carrier concentration n_i is given by:

$$n_i^2 = N_V N_C \exp(-E_{gap}/kT) \quad (25)$$

with

N_V = effective density of states in the valence band

N_C = effective density of states in the conduction band

E_{gap} = band gap energy

Both N_V and N_C are temperature dependent but this dependence is negligible compared to the exponential dependence of ni. When plotted in the Arrhenius diagram ($\ell n\ n_i$ versus l/T) the intrinsic range leads to a straight line with a slope - E/2k. At T~150K, the material containing dislocations turns extrinsic, i.e., the concentration of free charge carriers is given by the residual acceptor impurities, in this particular case shallow aluminum centers. These shallow centers, located -11 meV above the valence band remain ionized down to T-15K. Below this temperature the free holes become increasingly trapped by the negatively charged acceptor; one says the acceptors "freeze-out" or become neutral. The equilibrium between neutral acceptors A^n, ionized acceptors A' and free holes h can be described with a dissociation equation:

$$A^n \underset{\text{dissociation}}{\overset{\text{recombination}}{\rightleftharpoons}} A^- + h \quad (26)$$

It follows that

$$\frac{[A^-][h]}{[A^n]} = f(T) \propto \exp(-E_A/kT) = N_V \exp(-E_A/kT) \quad (27)$$

Assuming that there are no compensating donors present (an idealized case! not valid for Fig, 10), we can write:

$$[h] = [A^-] \tag{28}$$

Inserting eq.(23) in eq.(22) leads to a free hole ooncentratlon p:

$$p = [h] = (N_A N_V)^{1/2} \exp(-E/2kT) \tag{29}$$

for the case where $[h] \ll [A^n] + [A^-] = N_A$.

N_A = total acceptor concentration.

We predict that the logarithm of hole concentration drops linearly with the inverse of the absolute temperature.

$$\ln(p_1/p_2) = -2k\ EA\ \left(\frac{1}{T_1} - \frac{1}{T_2}\right) \tag{30}$$

As in the case of the intrinsic region we obtain a straight line in an Arrhenius plot.

In every "real" p-type semiconductor crystal, one finds a finite concentration of residual shallow donors N_D. When the free hole concentration drops, upon cooling, to values below N_D, we find that Eq.(28) is not valid any longer, The number of ionized acceptors will be equal to the number of the compensating residual donors. Appropriate modification of Eq.(29) leads to an expression with an exponential part $\exp(-E_A/kT)$ i.e., the slope of the hole freeze out curve doubles. The example shown In Fig. 11 only exhibits this latter regime because in ultrapure crystals most residual impurities, which form acceptors or donors, are present in comparable concentrations. This means that a freeze-out described by Eq.(29) cannot be clearly observed.

Figure 12 shows an Arrehnius plot of the free hole concentration as a function of the inverse absolute temperature of a gallium doped

silicon crystal: Such crystals are used for infrared sensors (photo conductors). The Hall effect measurements do not extend into the intrinsic range. Figure out the binding energy of the gallium acceptors, their concentration and estimate the residual donor concentration.

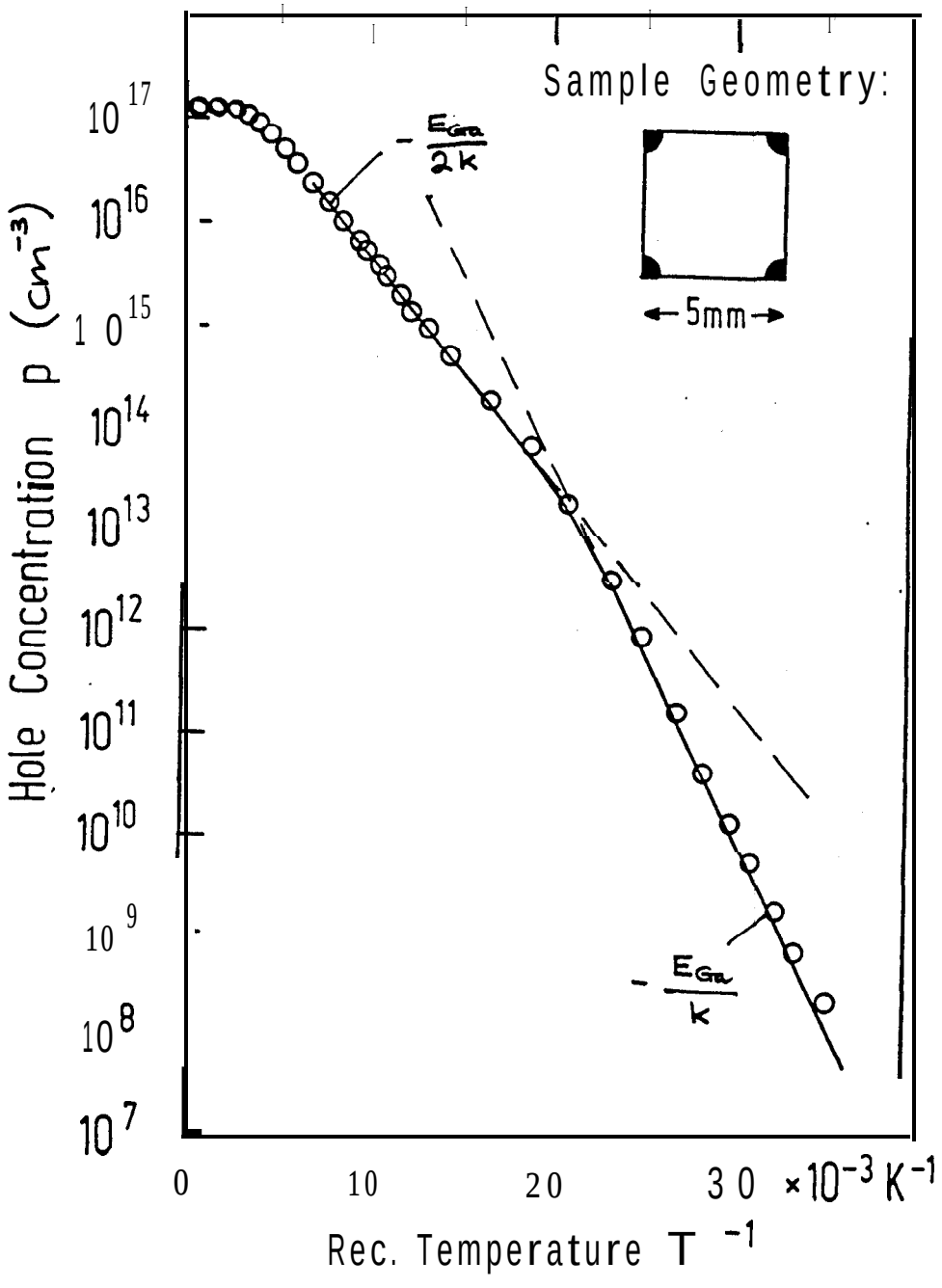


Figure 12